

Pain points of traditional development and how AI helps

While I hesitate to call it traditional development, there is a clear and significant shift in how work is done. Key pain points AI is addressing include:

1. Manual debugging. Debugging has always been time-consuming. Tools like GitHub Copilot now highlight errors in real time, suggesting fixes instantly.

2. Repetitive coding. Starting projects often meant rewriting boilerplate code. Now, a single line or keyword can trigger AI to suggest entire code blocks, saving significant time.

3. Testing. Unit testing used to be a burden. Today, AI generates test cases and testing frameworks, reducing repetitive effort.

4. Onboarding new developers. In large organisations, locating project files across platforms like SharePoint, GitHub, or ServiceNow can be overwhelming for new hires. AI-powered chatbots or plugins centralise knowledge, making onboarding faster and smoother.

5. Skill gaps. This is something I experienced firsthand. I once joined a project built in Golang, even though my core expertise was in Java. Initially, I had no idea where to start. GitHub Copilot became my guide, explaining the workflow, clarifying syntax, and even breaking down code line by line. It not only got the work done but also helped me learn Golang in the process.

GitHub Copilot: An AI pair programmer

Let's talk specifically about GitHub Copilot, which I have personally used extensively.

We do not call it an AI programmer. Instead, it is an AI pair programmer—a collaborator that supports, but does not replace, human creativity. Here are some

ways it has transformed my work:

- **Fixing code in-line.** Copilot highlights errors (with its familiar yellow star icon) and suggests corrections instantly

- **Understanding new code.** Perfect for working with unfamiliar languages or legacy projects

- **Unit testing and documentation.** Copilot auto-generates drafts, saving hours of manual effort

- **PR summaries and repo interaction.** It can generate pull request summaries and even “talk” to repositories

- **Error handling.** When errors appear in the terminal, Copilot immediately suggests possible fixes

One note of caution. Always ensure that sensitive code or data is not shared with these tools. Many organisations are rightly concerned about data security in AI-assisted workflows.

Final thoughts

AI is no longer just an accessory in software development, it is becoming a collaborator. From debugging and testing to onboarding and documentation, tools like GitHub Copilot are easing pain points that developers have lived with for years. But while the benefits are clear, the central point is that AI should add genuine value. Whether it is improving developer productivity or enhancing the end-user experience, the goal must always be impact, not hype.

As developers, we should embrace AI as a thought partner, a co-pilot, and a catalyst for innovation, while also being mindful of its limitations and risks. The journey from ‘writing code’ to ‘intelligent collaboration’ has just begun, and I am excited to see where it is taking us. **EFY**

The article is based on the talk titled ‘Transforming Software Development with AI’ given by Harsh Sharma (Shell) at AIDevCon 2025. It has been transcribed and developed by Vidushi Saxena, a journalist at EFY.



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